Universidade do Minho
Escola de Engenharia

Cooperative Tradeoff Analysis of Consortia for Plant-Based Fermentation

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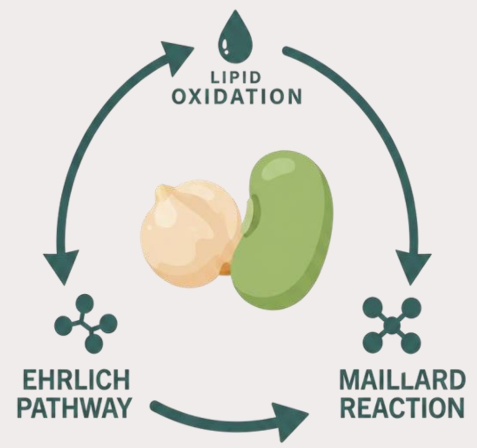
Introduction & Background

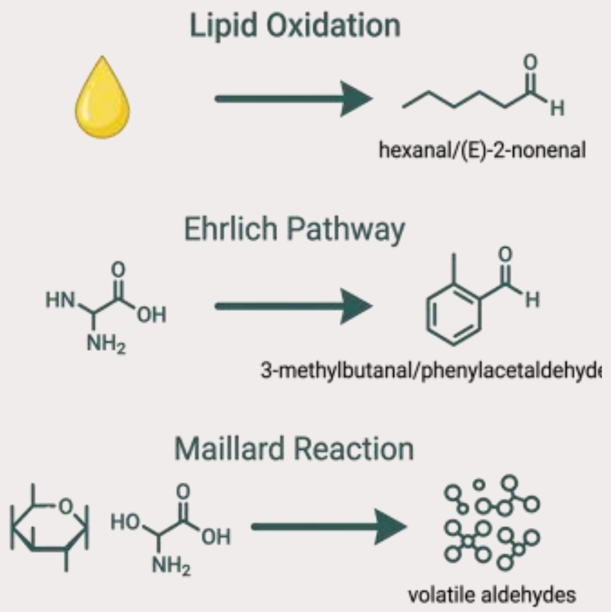
Chickpea and fava bean have strong nutritional profiles - high protein, fibre, micronutrients > candidates for animal protein substitution.

Problem : persistent **off-flavors** such as beany and grassy notes > limit consumer acceptance.

Three main pathways:

- lipoxygenase-mediated lipid oxidation originates volatile aldehydes such as hexanal and (E)-2 nonenal;
- protein degradation and amino acid catabolism through the Ehrlich pathway and Strecker degradation;
- accumulation of other volatile compounds from the Maillard reaction.





Motivation & Literature Gap

- Conventional mitigation of off-flavors relies on **empirical screening** - costly, biologically variable, and yields no mechanistic understanding. It identifies what works, but not why - particularly limiting when the goal is rational design rather than trial and error optimization.

- **Multi strain consortia**: offers metabolic complementarity and resilience, but cross-feeding and substrate competition produce outcomes that cannot be predicted from single-strain behavior:
    - A consortium of 2-3 LAB strains has a combinatorial interaction space that empirical methods cannot efficiently explore.

Rational consortium design remains an open challenge.

No published pipeline integrates genome-scale metabolic models into a nutritionally faithful legume fermentation context to predict both microbial interactions and off-flavor mitigation capacity.

LABs
as a biological solution

Contribution of LABs to the mitigation of off-flavors:

- **Acidification**: pH drop below 6.0–6.4 inhibits plant lipoxygenase;
- **Enzymatic action**: alcohol dehydrogenases convert volatile aldehydes to less odorous alcohols and acids.

L. plantarum WCFS1: has documented ADH activity in legume matrices.

L. mesenteroides: saccharolytic profile, heterofermentative, volatile aldehyde reduction capacity.

+

GEMs
as a computational tool

Genome-Scale Metabolic Models:

- reconstruct the full metabolic network
- allow computational prediction of fluxes, growth rates and metabolite production

FBA predicts growth rates and metabolite fluxes under defined constraints.

For small consortia with high-quality models, steady-state approaches achieve the best trade-off between predictive accuracy and computational cost (Scott et al., 2022).

Published curated GEMs available: *L.plantarum WCFS1* (iLP728) and *L.mesenteroides* (Koduru 2022).

Objectives

1. Curation and standardization of GEMs of iLP728 and Koduru 2022 GEMs;
2. Definition of medium conditions representative of legume matrices (chickpea and fava bean) in COBRApy;
3. Analysis of potential metabolic interactions and cross-feeding in the consortium using SMETANA;
4. Implementation of steady-state community simulations with MICOM to predict metabolic fluxes and off-flavor mitigation.

Methodology

Step 1/2 - Gem curation & Legume Medium Definition

Quality Analysis : memote score, reactions, genes, metabolites, growth & biomass spills

iLP728 - overly restrictive exchange bounds for auxotrophic strain > zero growth.

Curation made:

- necessary exchange reactions opened
- Added: GALt and RAFFINt transporters (GPR: lp_2732, lp_0843)

Koduru2022 - zero growth caused by single metabolite ID mismatch.

Curation made:

- correction of ID mismatch
- Namespace harmonized to BiGG convention

From composition to flux bounds:

- Nutritional data sourced from databases (FRIDA, USDA FoodData) and scientific articles;
- Convertion from mg/kg to mmol/gDW/h exchange bounds:

Sauer batch-culture equation:

$$qS \approx S_0 / (X \times t)$$

Category	Efficiency
Free sugars	1.00
Disaccharides	0.90
Amino acids	0.25
Raffinose	0.40/0.60
Lipids	0.50

Raw composition values are not equal to bioavailable substrate - organism-specific efficiency fractions are required to produce biologically meaningful flux bounds.

Methodology

Step 3 - SMETANA Interaction Analysis

SMETANA performs static analysis of potential metabolic exchanges between species before running full community simulations. It computes three interaction scores to identify likely cross-feeding interactions at low computational cost.

SC (Species Coupling):
Probability that one species requires another to grow. SC = 1.0 means topological dependency in the merged model.

MUS (Metabolite Uptake):
Probability that a species takes up a given metabolite from the shared pool. Identifies facultative cross-feeding capacity.

MPS (Metabolite Production):
Probability that a species produces a given metabolite into the shared pool. Identifies secretion capacity.

SMETANA results directed the selection of metabolites to monitor in MICOM - from static interaction space to targeted community simulation.

Methodology

Step 4 - MICOM Community Simulation

MICOM implements a two-level optimization approach particularly suitable for predicting cross-feeding and stable relative abundances in microbial consortia.

LEVEL 1: Maximise total community growth rate across all members.

LEVEL 2: Fix community growth ≥ fraction × maximum. Maximise individual member growth within that constraint.

Cooperative tradeoff fraction = 0.5

Three configurations tested in each legume matrix - chickpea and fava bean:

1. Abundance ratio: *L.plantarum* **75%** / *L.mesenteroides* **25%**
2. Abundance ratio: *L.plantarum* **50%** / *L.mesenteroides* **50%**
3. Abundance ratio: *L.plantarum* **25%** / *L.mesenteroides* **75%**

Off-flavor catalogue:

- 9 compounds tracked: hexanol, nonenol, nonenal, 2-methylbutanoate, 3-methylbutanoate, 2-methylpropanoate

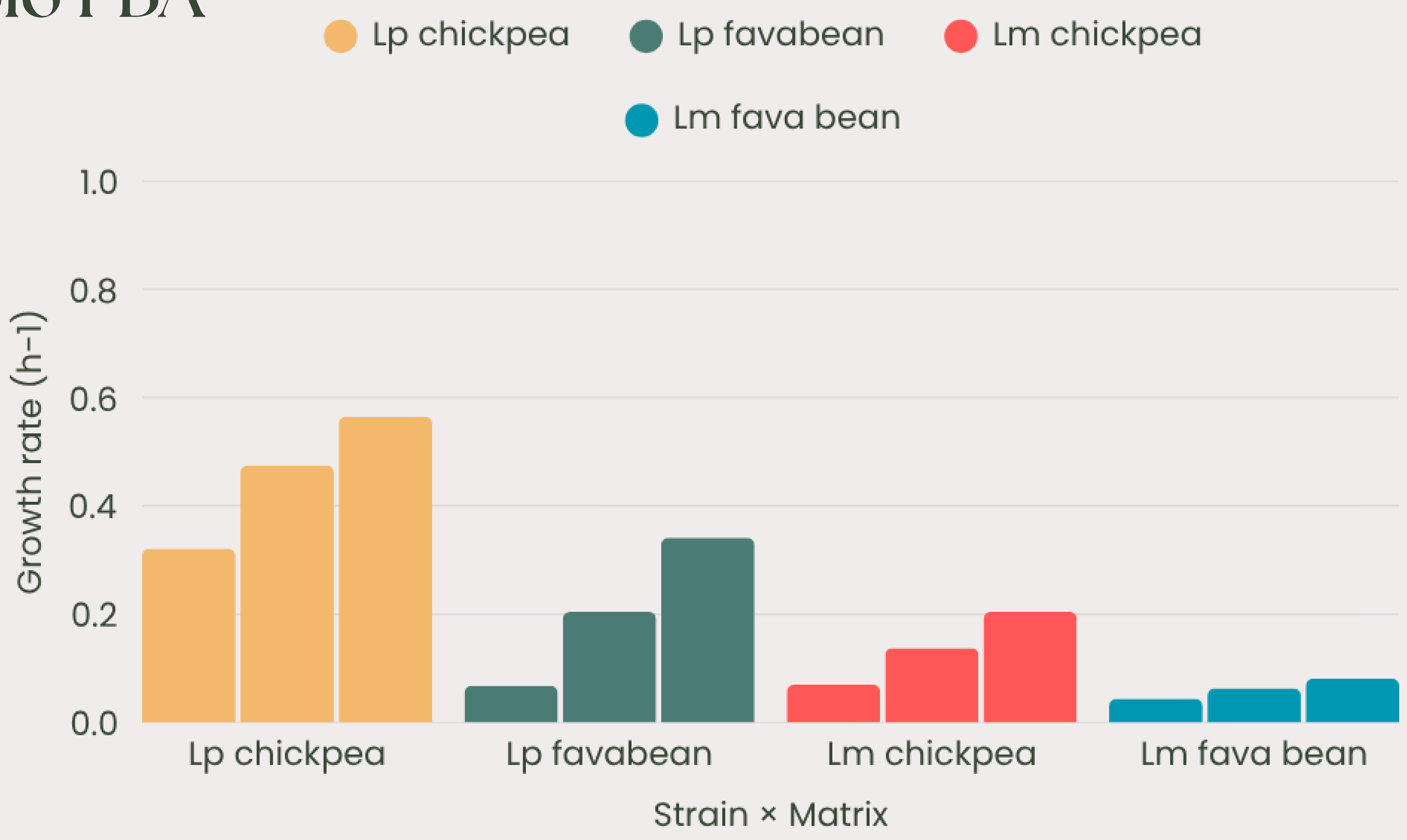
Steady-state simulation: no temporal dynamics. Strain-differential responses assessed via solo FBA perturbation analysis.

Results

Solo FBA

Each strain was optimized independently on the legume medium across three nutritional composition scenarios (min, median, max):

- *L. plantarum* (Lp) grows significantly faster in chickpea than in fava bean - reflects its dependence on disaccharides as primary carbon source.
- *L. mesenteroides* (Lm) grows comparably in chickpea, but is strongly limited in fava bean - lower free amino acid availability.



Perturbation analysis under solo FBA conditions reveals strain-differential metabolic responses:

- *L.plantarum* is primarily carbon-limited - growth drop with removal of sucrose/maltose; no sensitivity to amino acid or oxygen availability.
- *L.mesenteroides* is influenced by all three perturbations - broader nutritional dependencies; more versatile but less efficient metabolic profile.

Perturbation	Δ Lp	Δ Lm
No sucrose / maltose	-27.5%	-12.9%
Low amino acids	0.0%	-5.9%
Anaerobic	0.0%	-23.1%

Results

SMETANA Analysis

SMETANA computed three interaction scores across six nutritional scenarios - chickpea and fava bean × min, median, max.

- SC = 1.0 for *L. plantarum* as receiver across all scenarios - artefact of the merged model, not biological obligate dependency.
- MIP higher in chickpea (8–9) than fava bean (5–6) - chickpea creates more nutritional dependencies between strains, suggesting stronger potential for cross-feeding.
- MRO stable across scenarios (0.27–0.40) - moderate resource overlap confirms competition exists but does not dominate the interaction space.

Scenario	MIP	MRO	SC Lp→Lm	SC Lm→Lp	MUS > 0
Chickpea min	9	0.40	1.0	0.0	31
Chickpea med	8	0.40	1.0	0.0	31
Chickpea max	8	0.38	1.0	0.0	33
Fava bean min	5	0.27	1.0	0.0	24
Fava bean med	6	0.38	1.0	0.0	32
Fava bean max	6	0.35	1.0	0.0	36

Off-flavour acids (2-methylpropanoate, 2-methylbutanoate, 3-methylbutanoate) show consistent MU scores of 0.01–0.04 for *L. plantarum* across all scenarios, confirming facultative uptake capacity.

Results

MICOM Community Simulation

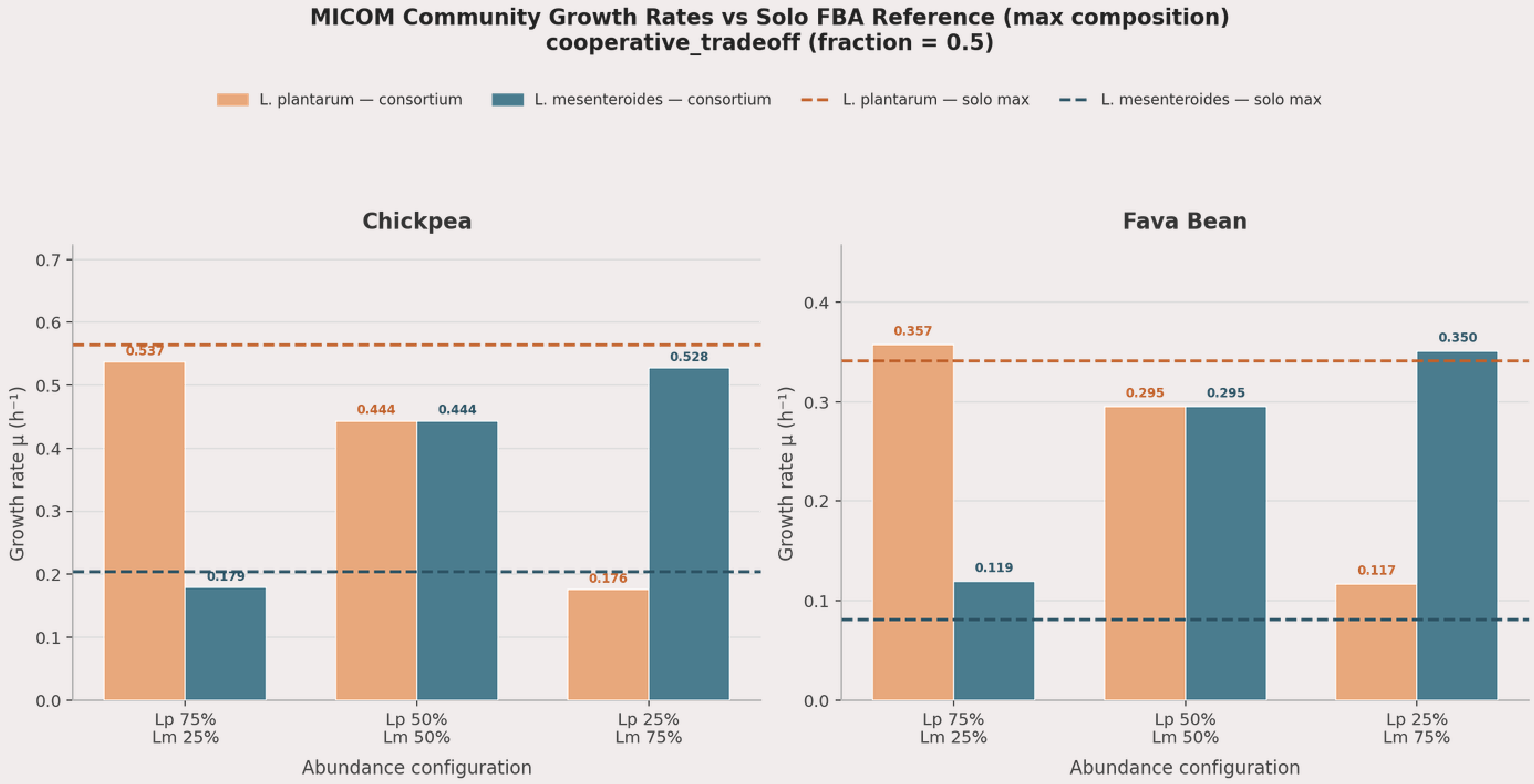
cooperative_tradeoff (fraction=0.5) run across 3 abundance configurations × 2 matrices :
6/6 configurations solved optimally

Chickpea:

- The consortium consistently benefits *L. mesenteroides*
- *L. plantarum* is penalized in all chickpea configurations - substrate competition for disaccharides

Fava bean:

- *L. mesenteroides* consistently benefits from the consortium
- *L. plantarum* benefits slightly in Lp75, the only configuration where it gains from the consortium.



Growth proportional to abundance in all configurations - structural consequence of MICOM abundance-scaled metabolic capacity.

Results

Cross-Feeding & Off-Flavour Mitigation Potential

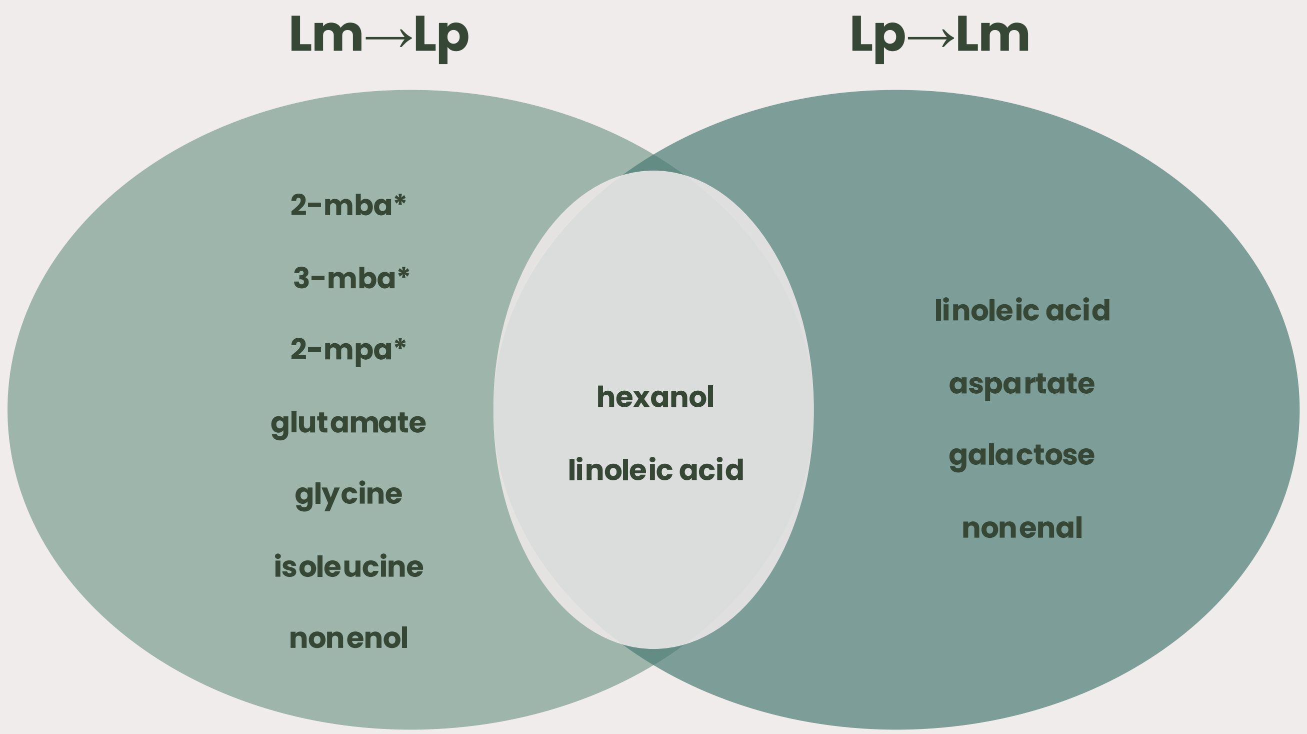
40 cross-feeding interactions detected consistently across abundance ratios and matrices

- *L. mesenteroides* secretes Ehrlich pathway acids and *L. plantarum* takes them up.
- Both strains consume linoleic acid → reduces LOX substrate → indirect hexanal limitation.
- *L .mesenteroides* converts Ehrlich aldehydes internally → secretes acids, not aldehydes → no volatile aldehyde accumulation in medium.

Direct ADH-mediated aldehyde conversion is not captured under the growth-maximizing FBA objective. Solo pFBA confirms production capacity exists.

- Quantitative prediction requires ATP maintenance constraints or dynamic simulation.

This pipeline establishes the metabolic interaction map necessary for off-flavor mitigation prediction - the foundation for future experimental validation.



*predicted by SMETANA and confirmed by MICOM

Conclusion & On-Going work

✓ Two curated, community-ready GEMs - iLP728 (*L. plantarum* WCFS1) and Koduru 2022 (*L. mesenteroides*) - corrected from zero growth to biologically plausible rates.

✓ A nutritionally parameterized legume medium pipeline applicable to any future LAB strain, converting compositional data into COBRApy exchange bounds via the Sauer equation.

✓ SMETANA-based characterization of the interaction space across six nutritional scenarios - facultative cross-feeding capacity confirmed for Ehrlich pathway acids.

✓ Functional MICOM community simulation across 6 configurations × 2 matrices, with 40 cross-feeding interactions detected and SMETANA predictions validated.

- ATP maintenance constraints to recover Ehrlich pathway aldehyde fluxes - enabling quantitative off-flavor prediction within the current model.
- Perturbation analysis extended to fava bean matrix - completing the strain-differential response characterization across both matrices.
- Formal documentation of solo pFBA Ehrlich fluxes - establishing production capacity baseline for future experimental validation.

Thank You

*No AI tools were used in the preparation or design of this presentation.